Requirement ID	Requirement Description	Source (PS / Stakeholder)	SRS Section	Related Model / Feature	Verification Method
External Interface Requirements					
FR-UI-01	The system shall be fully functional and visually consistent across the latest stable versions of Google Chrome, Mozilla Firefox, and Apple Safari.	Stakeholder (Usability)	3.1. User Interfaces	User Interface	Demonstration / Test
FR-UI-02	The interface shall employ a "Mobile First" responsive design, supporting screen widths ranging from 375px (mobile) to 1920px (desktop).	Stakeholder (Usability)	3.1. User Interfaces	User Interface	Demonstration / Test
FR-UI-03	All UI components shall adhere to WCAG 2.1 Level AA standards to support elderly users and those with visual impairments.	Stakeholder (Accessibility)	3.1. User Interfaces	User Interface	Inspection / Test
FR-UI-04	System navigation shall ensure that core features are reachable within a maximum of three (3) clicks from the main dashboard.	Stakeholder (Usability)	3.1. User Interfaces	User Interface	Demonstration / Test
FR-HWI-01	The application shall be deployed on secure, scalable cloud infrastructure (e.g., AWS or Azure) with data residency in Singapore.	Project Specification (CO-6)	3.2. Hardware Interfaces	Infrastructure	Inspection
FR-HWI-02	The system shall require a minimum of 4GB RAM and a modern processor for optimal performance on clinic-provided computers.	Stakeholder (IT Ops)	3.2. Hardware Interfaces	Infrastructure	Inspection
FR-HWI-03	The system shall interface with network printers for the generation of physical appointment slips and payment receipts.	Stakeholder (Clinic Ops)	3.2. Hardware Interfaces	Appointment Booking	Demonstration / Test
FR-SI-SP-01	The system shall integrate with the GovTech SingPass OIDC API to facilitate secure identity verification.	Project Specification (DE-3)	3.3. Software Interfaces	User Registration & Auth.	Demonstration / Test

FR-SI-DB-01	The system shall interface with a Relational Database (e.g., MySQL) for ACID-compliant storage of patient records.	Project Specification (CO-3)	3.3. Software Interfaces	Data Layer	Inspection
FR-SI-NTF-01	The system shall interface with the Twilio REST API for SMS alerts and the Amazon SES API for emails.	Project Specification (DE-2)	3.3. Software Interfaces	Notification Management	Demonstration / Test
FR-SI-SEC-01	The system shall integrate with a Web Application Firewall (WAF) to mitigate common OWASP Top 10 vulnerabilities.	Stakeholder (Security)	3.3. Software Interfaces	Security	Inspection
FR-CPI-01	All data transmitted between the client and server must be encrypted using TLS 1.2 or higher via HTTPS.	Project Specification (CO-5)	3.4. Communication Protocols	Security	Inspection / Test
FR-CPI-02	All external software interfaces shall utilize RESTful architectural style with JSON data payloads.	Project Specification (Internal)	3.4. Communication Protocols	Integration Layer	Inspection
FR-CPI-03	The system shall implement an exponential backoff retry mechanism for external API calls during service timeouts.	Stakeholder (Reliability)	3.4. Communication Protocols	Integration Layer	Test
System Features					
FR-AUTH-01	Users shall be able to create a new account by providing a valid email address and mobile number.	Project Specification	4.1. User Registration & Auth.	User Registration & Auth.	Demonstration / Test
FR-AUTH-02	Users shall be able to view their profile information including email, mobile number, and account creation date.	Stakeholder (Patients)	4.1. User Registration & Auth.	User Registration & Auth.	Demonstration / Test
FR-AUTH-03	Users shall be able to update their profile information including mobile number and password.	Stakeholder (Patients)	4.1. User Registration & Auth.	User Registration & Auth.	Demonstration / Test
FR-AUTH-04	Users shall be able to request account deactivation. Administrators shall have the ability to permanently delete user accounts in compliance with data retention policies.	Stakeholder (Compliance)	4.1. User Registration & Auth.	User Registration & Auth.	Demonstration / Test

FR-AUTH-05	Passwords shall contain a minimum of 8 characters, at least one uppercase letter, one lowercase letter, one number and one special character.	Stakeholder (Security)	4.1. User Registration & Auth.	User Registration & Auth.	Test
FR-AUTH-06	The system may support optional registration and login via SingPass.	Project Specification (DE-3)	4.1. User Registration & Auth.	User Registration & Auth.	Demonstration / Test
FR-AUTH-07	All login attempts shall be logged.	Stakeholder (Security)	4.1. User Registration & Auth.	User Registration & Auth.	Inspection / Test
FR-AUTH-08	The system shall lock a user account after five consecutive failed login attempts and require password reset or administrator verification.	Stakeholder (Security)	4.1 User Registration & Authentication	Account Security / Brute Force Protection	Test
FR-AUTH-09	User sessions shall automatically expire after 15 minutes of inactivity and require re-authentication.	Stakeholder (Security / Compliance)	4.1 User Registration & Authentication	Session Management	Test
FR-RBAC-01	Administrators shall be able to create new roles with custom permission sets and assign roles to users.	Stakeholder (Admin)	4.2. Role Based Access Control	Role Based Access Control	Demonstration / Test
FR-RBAC-02	The system shall read the user's role upon login and display the appropriate dashboard and menu options.	Project Specification	4.2. Role Based Access Control	Role Based Access Control	Demonstration / Test
FR-RBAC-03	Administrators shall be able to modify role permissions and reassign users to different roles.	Stakeholder (Admin)	4.2. Role Based Access Control	Role Based Access Control	Demonstration / Test
FR-RBAC-04	Administrators shall be able to deactivate roles ensuring no active users are assigned to deleted roles.	Stakeholder (Admin)	4.2. Role Based Access Control	Role Based Access Control	Test
FR-RBAC-05	Patients shall see booking options, appointment history, and profile management.	Project Specification	4.2. Role Based Access Control	Role Based Access Control	Demonstration / Test
FR-RBAC-06	Nurses shall manage appointment schedules, patient check-in/payment, and basic medical/prescription data.	Project Specification	4.2. Role Based Access Control	Role Based Access Control	Demonstration / Test
FR-RBAC-07	Doctors shall see their appointment schedules, patient lists, and have access to in-depth medical records.	Project Specification	4.2. Role Based Access Control	Role Based Access Control	Demonstration / Test

FR-RBAC-08	Administrators shall access clinic management, reporting, and user oversight panels.	Project Specification	4.2. Role Based Access Control	Role Based Access Control	Demonstration / Test
FR-BOOK-01	Patients shall be able to create a new appointment booking by selecting an available doctor, date, and time slot.	Project Specification	4.3. Appointment Booking & Mgmt	Appointment Booking & Mgmt	Demonstration / Test
FR-BOOK-02	Patients shall view their upcoming and past appointments including details. Staff can view all appointments based on permissions.	Stakeholder (Patients/Staff)	4.3. Appointment Booking & Mgmt	Appointment Booking & Mgmt	Demonstration / Test
FR-BOOK-03	Patients shall be able to reschedule appointments >24h in advance. Staff can update appointment status.	Stakeholder (Patients/Staff)	4.3. Appointment Booking & Mgmt	Appointment Booking & Mgmt	Demonstration / Test
FR-BOOK-04	Patients shall be able to cancel appointments >24h in advance. Staff can cancel at any time.	Stakeholder (Patients/Staff)	4.3. Appointment Booking & Mgmt	Appointment Booking & Mgmt	Demonstration / Test
FR-BOOK-05	Patients shall view real-time availability of doctors, filtered by clinic and date.	Stakeholder (Patients)	4.3. Appointment Booking & Mgmt	Appointment Booking & Mgmt	Demonstration / Test
FR-BOOK-06	Patients shall be limited to one active appointment per clinic per day.	Stakeholder (Business Rule)	4.3. Appointment Booking & Mgmt	Appointment Booking & Mgmt	Test
FR-BOOK-07	When a patient selects an available time slot, it shall be temporarily reserved for 5 minutes.	Stakeholder (Concurrency)	4.3. Appointment Booking & Mgmt	Appointment Booking & Mgmt	Test
FR-BOOK-08	Patients can freely cancel or reschedule appointments more than 24 hours in advance.	Stakeholder (Patients)	4.3. Appointment Booking & Mgmt	Appointment Booking & Mgmt	Demonstration / Test
FR-BOOK-09	Patients shall not be able to cancel or reschedule appointments via the system within 24 hours of the appointment time.	Stakeholder (Business Rule)	4.3. Appointment Booking & Mgmt	Appointment Booking & Mgmt	Test
FR-NTF-01	The system shall automatically create and send notifications when triggered by specific events.	Stakeholder (Patients)	4.4. Notification Management	Notification Management	Demonstration / Test

FR-NTF-02	Administrators shall be able to view notification logs including recipient, type, status, and timestamp.	Stakeholder (Admin)	4.4. Notification Management	Notification Management	Demonstration / Test
FR-NTF-03	Administrators shall be able to update notification templates for different notification types.	Stakeholder (Admin)	4.4. Notification Management	Notification Management	Demonstration / Test
FR-NTF-04	Administrators shall be able to archive old notification logs according to data retention policies.	Stakeholder (Compliance)	4.4. Notification Management	Notification Management	Inspection / Test
FR-NTF-05	A confirmation notification shall be sent immediately after a booking is successfully completed.	Stakeholder (Patients)	4.4. Notification Management	Notification Management	Demonstration / Test
FR-NTF-06	A reminder notification shall be sent 24 hours before a scheduled appointment.	Stakeholder (Patients)	4.4. Notification Management	Notification Management	Demonstration / Test
FR-NTF-07	A cancellation notification shall be sent immediately to the patient and relevant clinic staff when an appointment is cancelled.	Stakeholder (Patients/Staff)	4.4. Notification Management	Notification Management	Demonstration / Test
FR-NTF-08	Patients whose appointments are impacted by a doctor's schedule change shall be notified immediately.	Stakeholder (Patients)	4.4. Notification Management	Notification Management	Demonstration / Test
FR-DOC-01	Doctors shall be able to create unavailability blocks. Administrators can create unavailability on behalf of doctors.	Stakeholder (Doctors/Admin)	4.5. Doctor Availability Mgmt	Doctor Availability Mgmt	Demonstration / Test
FR-DOC-02	Doctors shall view their current availability schedule. Patients shall read available time slots.	Stakeholder (Doctors/Patients)	4.5. Doctor Availability Mgmt	Doctor Availability Mgmt	Demonstration / Test
FR-DOC-03	Doctors shall be able to modify existing unavailability blocks at least 48 hours in advance. Within 48 hours, modifications require administrator approval.	Stakeholder (Doctors)	4.5. Doctor Availability Mgmt	Doctor Availability Mgmt	Test
FR-DOC-04	Doctors shall be able to remove unavailability blocks, provided no patients were affected.	Stakeholder (Doctors)	4.5. Doctor Availability Mgmt	Doctor Availability Mgmt	Demonstration / Test
FR-DOC-05	Doctors can block out personal time, provided the change is made at least 48 hours in advance.	Stakeholder (Doctors)	4.5. Doctor Availability Mgmt	Doctor Availability Mgmt	Test

FR-DOC-06	Any unplanned unavailability within 24 hours of an appointment must be approved by an administrator.	Stakeholder (Business Rule)	4.5. Doctor Availability Mgmt	Doctor Availability Mgmt	Test
FR-DOC-07	The system shall automatically identify affected patients and trigger the notification process.	Stakeholder (Patients)	4.5. Doctor Availability Mgmt	Notification Management	Test
FR-WL-01	Patients shall be able to join a waiting list for a specific doctor, date, and time slot.	Stakeholder (Patients)	4.6. Waiting List Management	Waiting List Management	Demonstration / Test
FR-WL-02	Patients shall be able to view their position in the waiting list. Staff shall be able to view the complete waiting list.	Stakeholder (Patients/Staff)	4.6. Waiting List Management	Waiting List Management	Demonstration / Test
FR-WL-03	The system shall automatically update waiting list positions when slots become available and when patients claim or decline offers.	Project Specification	4.6. Waiting List Management	Waiting List Management	Test
FR-WL-04	Patients shall be able to remove themselves from the waiting list at any time.	Stakeholder (Patients)	4.6. Waiting List Management	Waiting List Management	Demonstration / Test
FR-WL-05	Patients who attempt to book a fully booked slot may be given the option to join a waiting list on a first-come-first-serve basis.	Project Specification	4.6. Waiting List Management	Waiting List Management	Demonstration / Test
FR-WL-06	When a slot becomes available, the first patient on the waiting list shall be notified.	Stakeholder (Patients)	4.6. Waiting List Management	Waiting List Management	Test
FR-WL-07	The offered slot must be claimed within 2 hours. If not claimed, the system shall offer the slot to the next patient.	Stakeholder (Business Rule)	4.6. Waiting List Management	Waiting List Management	Test
FR-REF-01	Doctors shall be able to create a new referral letter by populating a template. The system shall create a referral record.	Stakeholder (Doctors)	4.7. Referral Processing	Referral Processing	Demonstration / Test
FR-REF-02	Authorized users shall be able to view referral letters based on their permissions.	Stakeholder (Staff/Patients)	4.7. Referral Processing	Referral Processing	Demonstration / Test
FR-REF-03	Doctors shall be able to update referral letters. Any update shall create a new version while preserving the original.	Stakeholder (Doctors)	4.7. Referral Processing	Referral Processing	Test

FR-REF-04	Completed referral letters cannot be permanently deleted. Administrators may archive a referral.	Stakeholder (Compliance)	4.7. Referral Processing	Referral Processing	Test
FR-REF-05	Nurses shall be able to provide referral letters given by doctors to patients, via print or secure electronic copy.	Stakeholder (Nurses)	4.7. Referral Processing	Referral Processing	Demonstration / Test
FR-REF-06	All referral letters shall be stored as part of the patient's permanent digital record.	Stakeholder (Compliance)	4.7. Referral Processing	Referral Processing	Inspection
Other Non-Functional Requirements					
NFR-PERF-01	All web pages shall load within 5 seconds under normal network conditions.	Stakeholder (Performance)	5.1. Performance Requirements	System Performance	Test
NFR-PERF-02	A complete booking transaction shall complete within 10 seconds.	Stakeholder (Performance)	5.1. Performance Requirements	System Performance	Test
NFR-PERF-03	The system shall support a minimum of 100 concurrent active users without performance degradation.	Stakeholder (Performance)	5.1. Performance Requirements	System Performance	Test
NFR-SAFE-01	The system shall prevent partial or corrupted appointment records from being saved.	Stakeholder (Reliability)	5.2. Safety Requirements	Data Integrity	Test
NFR-SAFE-02	In the event of a booking transaction failure, the system shall clearly notify the user and ensure no duplicate appointments.	Stakeholder (Reliability)	5.2. Safety Requirements	Appointment Booking	Test
NFR-SAFE-03	The system shall require explicit user confirmation before completing irreversible actions.	Stakeholder (Usability)	5.2. Safety Requirements	User Interface	Inspection / Test
NFR-SAFE-04	In the event of temporary service unavailability, clinic staff shall retain read-only access to existing appointment schedules.	Stakeholder (Ops Continuity)	5.2. Safety Requirements	System Resilience	Test
NFR-SEC-01	The system shall be designed and maintained in full compliance with Singapore's PDPA and MOH guidelines.	Project Specification (CO-4)	5.3. Security Requirements	Security & Compliance	Inspection
NFR-SEC-02	All sensitive patient data shall be encrypted at rest and in transit (TLS 1.2+).	Project Specification (CO-5)	5.3. Security Requirements	Security & Compliance	Inspection / Test

NFR-SEC-03	Audit logs shall track all access to sensitive patient data. All administrative actions shall be logged.	Stakeholder (Security)	5.3. Security Requirements	Security & Compliance	Inspection / Test
NFR-SEC-04	The system shall support configurable data retention and deletion policies in accordance with PDPA.	Stakeholder (Compliance)	5.3. Security Requirements	Security & Compliance	Demonstration / Test
NFR-SEC-05	Patient consent mechanisms shall be integrated into all relevant data collection points.	Stakeholder (Compliance)	5.3. Security Requirements	Security & Compliance	Inspection
NFR-SEC-06	Regular security testing must be conducted pre-launch and at least annually thereafter.	Stakeholder (Security)	5.3. Security Requirements	Security & Compliance	Inspection
NFR-AV-01	The patient booking portal shall achieve at least 99% uptime, measured monthly.	Stakeholder (Reliability)	5.4. Software Quality Attributes	System Performance	Inspection / Test
NFR-REL-01	The system shall maintain 100% data consistency with zero data loss in the event of a component failure.	Stakeholder (Reliability)	5.4. Software Quality Attributes	Data Integrity	Test
NFR-SCAL-01	The system architecture shall support the onboarding of additional clinic locations without requiring core system redesign.	Stakeholder (Scalability)	5.4. Software Quality Attributes	System Architecture	Inspection
NFR-MON-01	The system shall include centralized logging, monitoring, and alerting mechanisms.	Stakeholder (Maintainability)	5.4. Software Quality Attributes	System Maintenance	Inspection
NFR-MOD-01	Compliance logic shall be modularized to accommodate regulatory changes with minimal impact.	Stakeholder (Maintainability)	5.4. Software Quality Attributes	System Architecture	Inspection
Project & User Documentation					
PD-REQ-01	The vendor shall provide a Solution Architecture Diagram.	Stakeholder (Technical)	5.6. Project Documentation	Documentation	Inspection
PD-REQ-02	The vendor shall provide a High-Level Data Model.	Stakeholder (Technical)	5.6. Project Documentation	Documentation	Inspection
PD-REQ-03	The vendor shall provide monthly progress reports.	Stakeholder (Project Mgmt)	5.6. Project Documentation	Project Management	Inspection
UD-REQ-01	The vendor shall provide user manuals and quick-reference guides for all user roles.	Stakeholder (Training)	5.7. User Documentation	Documentation	Inspection
UD-REQ-02	The vendor shall provide training materials and conduct training sessions for clinic staff and administrators.	Stakeholder (Training)	5.7. User Documentation	Documentation / Training	Demonstration